

COMMON COURSE ASSIGNMENT FORMAT

A. Assignment Information

Particular	Details
Department:	B Tech CS and BIO SCI
Programme:	B.Tech
Course Code & Course Name	CSA0709&Computer Networks
Academic Year / Batch	3rd Year/2026
Faculty Name	Dr.Senthimal Selvan
Assignment Title	Design an FTP-based file-sharing network using Packet Tracer. Configure the FTP service and test communication between clients and server
Date of Issue	31-08-2026
Date of Submission	01-09-2026
Maximum Marks	100
Course Outcome(s) – CO	Design, simulate and evaluate computer network protocols and services using modern networking tools
Bloom's Taxonomy Level	Apply / Analyze (L3–L4)
SDG Mapping (SDG 1-17)	SDG 4 – Quality Education; SDG 9 – Industry, Innovation and Infrastructure
Industry / Societal Relevance	FTP-based file servers underpin enterprise data exchange, software distribution and DevOps pipelines; understanding client–server file protocols is foundational to network manageristration and cloud infrastructure careers

B. Assignment Problem / Challenge

This assignment requires the design, configuration and validation of a small enterprise-style local area network in which a dedicated File Transfer Protocol (FTP) server provides centralized file storage and retrieval services to multiple client workstations, using Cisco Packet Tracer as the simulation environment.

The task requires the student to apply course concepts of client–server protocols, TCP connection management and network addressing, rather than reproduce theory; to identify and analyze the design problem of secure, multi-user file sharing; to consider performance, security and access-control constraints; to configure the FTP service using a modern networking simulation tool; to interpret the results of live FTP sessions; and to justify the design and configuration decisions made.

C. Problem Statement

Problem / Case / Design Challenge:

A small academic computing unit needs a centralized system through which users on different workstations can upload, download, list and manage shared files from one server. Design and implement an FTP-based file-sharing network in Cisco Packet Tracer consisting of one FTP server and three client PCs connected through a switch. Configure the FTP service with role-based user accounts, assign a consistent static IP addressing scheme, and test connectivity, authentication, directory listing, upload, download and deletion operations to verify reliable end-to-end communication.

D. Requirements and Constraints

The following requirements and constraints, drawn from the general list in the assignment format, are relevant to this networking design problem:

1. Functional requirement: clients must be able to connect to the server, authenticate, list directory contents, upload and download files.
2. Performance requirement: the network must support at least three simultaneous client sessions without connection failure.
3. Technical constraint: all devices must reside on a single Class C private subnet (192.168.10.0/24) with static IP addressing.
4. Resource constraint: only Cisco Packet Tracer's built-in Server, Switch and PC objects and their native Services tab are used — no external FTP daemon.
5. Security/privacy: access must be authenticated (username/password); a visitor account with reduced privileges illustrates role-based access control.
6. Reliability: FTP runs over TCP, so the design must rely on TCP's connection-oriented, acknowledged delivery for data integrity.

7. Standards/codes: FTP configuration follows RFC 959 (File Transfer Protocol) semantics as implemented in Packet Tracer's Server service.

E. Student Work

1. Problem Understanding and Formulation

Problem: Students and staff working from separate client machines need a common, reliable location to store and retrieve academic files instead of relying on ad-hoc transfers. A dedicated file server accessible over the LAN using a standard file-transfer protocol addresses this need.

Expected outcomes: a functioning topology in which every client can reach the server, authenticate with distinct credentials, and successfully list, upload, download and (where permitted) delete files, verified through the Packet Tracer command prompt and simulation view.

Information/data available: Packet Tracer's device library (Server-PT, 2960 Switch, PC-PT), the Services tab exposing the FTP service with per-user permission controls, and the command-prompt FTP client available on each PC object.

Assumptions made: all devices are newly deployed with no prior configuration; the network is isolated and does not require Internet access; FastEthernet links are reliable; and the academic unit is small enough for a single flat subnet without VLAN segmentation or a router.

Constraints to satisfy: static addressing within 192.168.10.0/24, authenticated access, differentiated permissions per account, and successful two-way file transfer confirmed for every client.

2 . Application of Course Knowledge

The design applies the client–server model and TCP/IP protocol suite studied in the course. FTP is an application-layer protocol that uses TCP and separates control and data communication: the control connection uses well-known port 21, while the data connection is used for directory listings and file transfers.

Each client–server exchange therefore follows the standard TCP three-way handshake (SYN, SYN-ACK, ACK) before any FTP command is accepted, and address resolution (ARP) precedes the handshake so that the client's default gateway-less, same-subnet frame can be delivered at the data-link layer using the server's MAC address.

Addressing follows classless design principles: the 192.168.10.0/24 private address block provides 254 usable host addresses, which is more than sufficient for one server and three clients. The mask 255.255.255.0 is applied uniformly. Because all devices share one broadcast domain, no default gateway is functionally required for local communication.

3. Solution / Design / Methodology

Two candidate topologies were considered before finalizing the design.

- Option A — Flat single-switch LAN: Server and all clients connect to one Cisco 2960 switch; addressing and configuration are minimal and suitable for a small academic lab. (Selected.)
- Option B — Routed two-subnet LAN: clients on one subnet and the server on another, joined by a router; more realistic for a larger organization but adds routing configuration that is outside the scope of this problem.

Option A was selected because the problem focuses on local client-to-server file sharing and does not require inter-departmental segmentation. The flat topology reduces configuration complexity and keeps the main focus on FTP service operation.

Final Topology

Devices: 1 × Server-PT (FTP server), 1 × Cisco 2960 switch and 3 × PC-PT clients. All devices are connected using straight-through copper Ethernet cables on FastEthernet ports.

Device	Interface	IP Address	Subnet Mask	Default GW
FTP Server (Server0)	FastEthernet0	192.168.10.10	255.255.255.0	—
Client PC1	FastEthernet0	192.168.10.11	255.255.255.0	192.168.10.10
Client PC2	FastEthernet0	192.168.10.12	255.255.255.0	192.168.10.10
Client PC3	FastEthernet0	192.168.10.13	255.255.255.0	192.168.10.10

On the server, the Services tab is opened and FTP is switched On. Three role-based accounts are created under the FTP configuration panel with individually selected permission checkboxes (Write, Read, Delete, Rename, List):

Username	Password	Permissions Granted
manager	mgr456	Read, Write, Delete, Rename, List
user1	user789	Read, Write, List

Username	Password	Permissions Granted
visitor	view321	Read, List

Pseudo code :

Pseudo code representing the logical sequence of an FTP client-server file-transfer session, corresponding to the commands executed from each client's command prompt in Packet Tracer:

BEGIN FTP_Session

client.connect(server_ip = "192.168.10.10", port = 21)

IF connection.status != SUCCESS THEN

 PRINT "Connection failed - check IP/gateway/cabling"

 EXIT

END IF

PRINT "Connected to 192.168.10.10"

username = INPUT("User: ")

password = INPUT("Password: ")

auth_result = server.authenticate(username, password)

IF auth_result != SUCCESS THEN

 PRINT "Login incorrect"

 EXIT

END IF

PRINT "230 Login successful"

LOOP

 command = INPUT("ftp> ")

 SWITCH command:

 CASE "dir":

```
    file_list = server.listFiles()
    PRINT file_list
CASE "put <filename>":
    IF user.permission INCLUDES "write" THEN
        server.storeFile(filename, client.readFile(filename))
        PRINT "226 Transfer complete"
    ELSE
        PRINT "550 Permission denied"
    END IF
CASE "get <filename>":
    IF user.permission INCLUDES "read" THEN
        data = server.retrieveFile(filename)
        client.saveFile(filename, data)
        PRINT "226 Transfer complete"
    ELSE
        PRINT "550 Permission denied"
    END IF
CASE "quit":
    client.disconnect()
    EXIT LOOP
DEFAULT:
    PRINT "Unknown command"
END SWITCH
END LOOP
END FTP_Session
```


FTP-Based File-Sharing Network

Design, configuration and testing using Cisco Packet Tracer



Subnet: 192.168.10.0/24 | Clients: 3 | Protocol: FTP over TCP

Network Topology

Static topology diagram of the FTP file-sharing network



All devices share one broadcast domain; no router is required for local communication.

IP Address Table

Static IP assignments for all network devices

DEVICE	IP ADDRESS	SUBNET MASK	GATEWAY
FTP Server	192.168.10.10	255.255.255.0	—
PC1	192.168.10.11	255.255.255.0	192.168.10.10
PC2	192.168.10.12	255.255.255.0	192.168.10.10
PC3	192.168.10.13	255.255.255.0	192.168.10.10

FTP Simulator — Network View

Three client PCs connected through SW1 to the FTP server



Network: 192.168.10.0/24 | Mask: 255.255.255.0

FTP User Accounts

Role-based permissions configured on the server

MANAGER

Username: manager

Password: mgr456

Read ✓ Write ✓ Delete ✓ Rename ✓ List ✓

USER1

Username: user1

Password: user789

Read ✓ Write ✓ Delete ✗ Rename ✗ List ✓

VISITOR

Username: visitor

Password: view321

Read ✓ Write ✗ Delete ✗ Rename ✗ List ✓

FTP Session — PC1

Logged in as manager with full permissions

FTP OPERATIONS

DIR

UPLOAD project.txt

DOWNLOAD notice.txt

DELETE project.txt

COMMAND PROMPT

```
ftp 192.168.10.10
User: manager
Password: *****
230 Login successful
ftp> dir
ftp> put project.txt
226 Transfer complete
ftp> get notice.txt
226 Transfer complete
```

Configuration & FTP Commands

Step-by-step setup in Cisco Packet Tracer

CONFIGURATION STEPS

1. Place Server-PT, 2960 switch and three PC-PT devices.
2. Connect all devices with copper straight-through links.
3. Assign 192.168.10.10-13 using /24 mask.
4. Enable Services → FTP on the server.
5. Create manager, user1 and visitor accounts.
6. Apply full, partial and read-only permissions.
7. Test ping, login, listing and file transfer.

FTP COMMANDS

```
ftp 192.168.10.10

dir

put project.txt

get notice.txt

delete project.txt

quit
```

Testing & Validation

Functional test cases after configuration

TEST ID	DESCRIPTION	EXPECTED RESULT	STATUS
TC-01	Ping FTP server	Reply received, 0% loss	PASS
TC-02	Valid FTP login	230 Login successful	PASS
TC-03	Invalid password	530 Login incorrect	PASS
TC-04	Directory listing	File list returned	PASS
TC-05	Upload project.txt	226 Transfer complete	PASS
TC-06	Download notice.txt	226 Transfer complete	PASS
TC-07	Visitor delete attempt	550 Permission denied	PASS
TC-08	Manager delete	250 File deleted	PASS
TC-09	PC2 + PC3 concurrent access	Sessions served independently	PASS

Key Networking Concepts

Protocols and roles used in the design

FTP

File Transfer Protocol — application-layer service for file exchange.

TCP

Reliable, connection-oriented transport protocol.

Client-Server Model

Clients request services from a centralized server.

Port 21

Control connection used for FTP commands and replies.

TCP Handshake

SYN → SYN-ACK → ACK establishes a TCP connection.

Port 20 / Data Channel

Data connection used during transfers in active mode.

FTP vs TFTP vs HTTP

Protocol comparison for centralized file sharing

PROTOCOL	TRANSPORT	KEY FEATURES	SUITABILITY
FTP	TCP	Authentication, directory listing, upload/download	Best fit
TFTP	UDP	No authentication; simple file transfer	Not suitable
HTTP	TCP	Web-based transfer; limited native file control	Possible

FTP Network Project Packet Tracer Lab

Security Considerations

Important limitations and good practices

Cleartext Credentials

Traditional FTP sends usernames and passwords without encryption.

Use SFTP / FTPS

For real-world deployment, prefer encrypted file-transfer protocols.

Educational Scope

This Packet Tracer design is intended for a controlled academic LAN.

FTP Network Project Packet Tracer Lab

Conclusion

Summary of the modified network implementation

The three-client FTP network was implemented on the 192.168.10.0/24 subnet using one Cisco 2960 switch and one FTP server. Manager, user1 and visitor accounts were configured with different permissions. All nine validation tests produced the expected results, confirming connectivity, authentication, file transfer and access control.

Overall result: 9/9 test cases passed

4. Use of Modern Tools

Use appropriate tools wherever relevant.

Examples:

CSE/IT: Python, MATLAB, TensorFlow, cloud platforms, Git, simulation tools

ECE/EEE: MATLAB/Simulink, Cadence, Synopsys, Vivado, Multisim, LTspice

Mechanical: ANSYS, SolidWorks, CATIA, MATLAB

Civil: STAAD.Pro, ETABS, AutoCAD, Revit

Biomedical: MATLAB, LabVIEW, signal/image-processing tools

Biotechnology: Bioinformatics/data-analysis/simulation tools

Students shall provide appropriate evidence of tool usage and outputs.

5 . Results and Validation

Each client was tested against the same sequence of operations — connectivity, authentication, directory listing, upload, download and deletion — with the results summarized below. All observed FTP reply codes match RFC 959 expectations (220 service ready, 331/230 for login, 226 for completed transfer, 550 for permission failures).

ID	Test Case	Expected / Observed Result	Status
TC-01	Ping from PC1 to Server (192.168.10.10)	Reply received, 0% loss	Pass
TC-02	FTP login with valid credentials (manager/mgr456)	331 Username OK, 230 Login successful	Pass
TC-03	FTP login with invalid password	530 Login incorrect	Pass
TC-04	List directory contents (dir)	File listing returned from server	Pass
TC-05	Upload file to server (put project.txt)	226 Transfer complete; file visible on server	Pass
TC-06	Download file from server (get notice.txt)	226 Transfer complete; file saved on client	Pass
TC-07	Delete file with visitor account (delete project.txt)	550 Access denied (no delete permission)	Pass (expected failure)
TC-08	Delete file with manager account	250 File deleted	Pass
TC-09	Simultaneous access — PC2 and PC3 connect while PC1 is transferring	Both sessions served independently	Pass

Validation against the stated requirements: connectivity (TC-01) confirms the addressing scheme is correctly applied; authentication (TC-02/TC-03) confirms credential enforcement; TC-04–TC-06 confirm the core functional requirement (list, upload, download); TC-07/TC-08 confirm role-based access control between the manager and visitor accounts; and TC-09 confirms the performance requirement of concurrent client sessions. All nine test cases produced the expected result, so the implemented network satisfies every requirement stated in Section D.

6. Analysis and Engineering Decision

FTP was compared against two alternative file-transfer approaches available conceptually within the same network:

Criterion	FTP	TFTP	HTTP
Connections used	Two (control – port 21, data – port 20/ephemeral)	One (port 69, UDP)	One (port 80/443, TCP)
Transport protocol	TCP (reliable)	UDP (unreliable)	TCP (reliable)
Authentication	Username / password	None	Optional (HTTP auth)
Typical use case	Bulk file transfer, directory browsing	Bootimg devices, firmware transfer	Web-based single-file download
Suitability for this assignment	Best fit — supports login, listing,	Not suitable — no authentication or	Possible but lacks native directory/user-

Criterion	FTP	TFTP	HTTP
	upload/download as required	directory browsing	management features

FTP was selected over TFTP because the problem requires authenticated, permission-differentiated access, which TFTP does not provide. FTP was selected over HTTP because the Packet Tracer FTP service directly provides user accounts, directory listing and file-management permissions, matching the assignment requirements more closely.

Trade-off noted: traditional FTP transmits usernames and passwords without encryption, which is a security weakness. A real deployment should use FTPS or SFTP, but encrypted FTP services are outside the scope of this Packet Tracer exercise.

7. Broader Considerations

- Security: cleartext authentication is a genuine risk if this design were deployed on a real, non-isolated network; production systems should use FTPS/SFTP or restrict FTP to a trusted internal segment.
- Professional responsibility: differentiated account permissions (manager vs. visitor) reflect the principle of least privilege, limiting accidental or unauthorized data loss.
- Accessibility/Society: a centralized file server reduces duplicated effort and version-conflict errors among users sharing documents, a common organizational need.

- Economics: the design uses only free simulation tooling and standard, license-free protocols, keeping the solution low-cost to prototype before any real hardware purchase.

8 . Conclusion

A three-client, single-server FTP file-sharing network was designed and implemented in Cisco Packet Tracer on the 192.168.10.0/24 subnet. The FTP service was configured with manager, user1 and visitor accounts having different permissions. All nine functional test cases covering connectivity, authentication, listing, upload, download, deletion and concurrent access produced the expected results. The principal limitation is FTP's cleartext credential exchange, so FTPS or SFTP would be preferred for real-world deployment.

9. Student Reflection

Beyond the taught theory of the client–server model, this exercise reinforced how addressing, authentication and file permissions work together in a practical network. I also learned how to validate an application service using multiple test cases instead of relying only on a successful ping. With additional time, the design could be extended to multiple subnets, ACLs and a more secure SFTP/FTPS implementation.

10 . References

1. **Cisco Networking Academy. Cisco Packet Tracer Tutorials and Resources.** Cisco, 2026. Available at: <https://www.netacad.com/courses/packet-tracerhttps://www.netacad.com/courses/packet-tracer>
2. **Cisco. Cisco Packet Tracer: Simulation and Network Visualization.** Cisco Learning Network, 2026. Available at: <https://learningnetwork.cisco.com/https://learningnetwork.cisco.com/>
3. **Kurose, J. F.,&Ross, K. W.** *Computer Networking: A Top-Down Approach*. 9th Edition, Pearson, 2025.
(Useful for FTP and application-layer communication.)
4. **Cisco Networking Academy. Networking Basics and Network Simulation Learning Materials.** Cisco, 2025. Available at: <https://www.netacad.com/>

Common Assessment Rubric

Assessment Criterion	Marks
Problem understanding & formulation	10
Application of course/domain knowledge	20
Solution methodology / design / implementation	20
Use of appropriate modern tools / techniques	10
Results, testing & validation	15
Analysis, trade-offs & justification	15

Broader considerations / professional responsibility, where applicable	5
Technical documentation & reflection	5
TOTAL	100

CO–PO–Assessment Mapping

Assessment Component	CO	PO(s)	Bloom's Level	Marks
Problem formulation			L3/L4	10
Application of knowledge			L3/L4	20
Solution / Design / Implementation			L4/L5/L6	20
Modern tool usage			L3/L4	10
Validation			L4/L5	15
Analysis & justification			L4/L5	15
Broader considerations			L4/L5	5
Documentation & reflection			L3/L4	5

Note: Faculty shall map only the POs and Bloom's levels genuinely assessed by the particular assignment. Every assignment is **not required to map to every PO** .

G.Faculty Evaluation & Continuous Improvement

CO Attainment

Performance Level	Criterion
Level 3	$\geq 70\%$

Level 2	60–69%
Level 1	50–59%
Level 0	< 50%

Target CO Attainment: _____

Actual CO Attainment: _____

Faculty Analysis

Areas in which students performed well:

Areas requiring improvement:

Corrective / Improvement Action:

Action to be implemented in the next offering:

Documents to be Maintained

The department/course faculty shall maintain:

- 1 . Assignment question/problem statement
- 2 . CO–PO and Bloom's mapping
- 3 . Assessment rubric
- 4 . Student submissions
- 5 . Tool/simulation/experimental evidence, where applicable
- 6 . Evaluated scripts/reports
- 7 . Marks and rubric-wise assessment
- 8 . CO attainment analysis
- 9 . Sample student work representing different performance levels

